

PH110-PHYSICS (Modern Physics)

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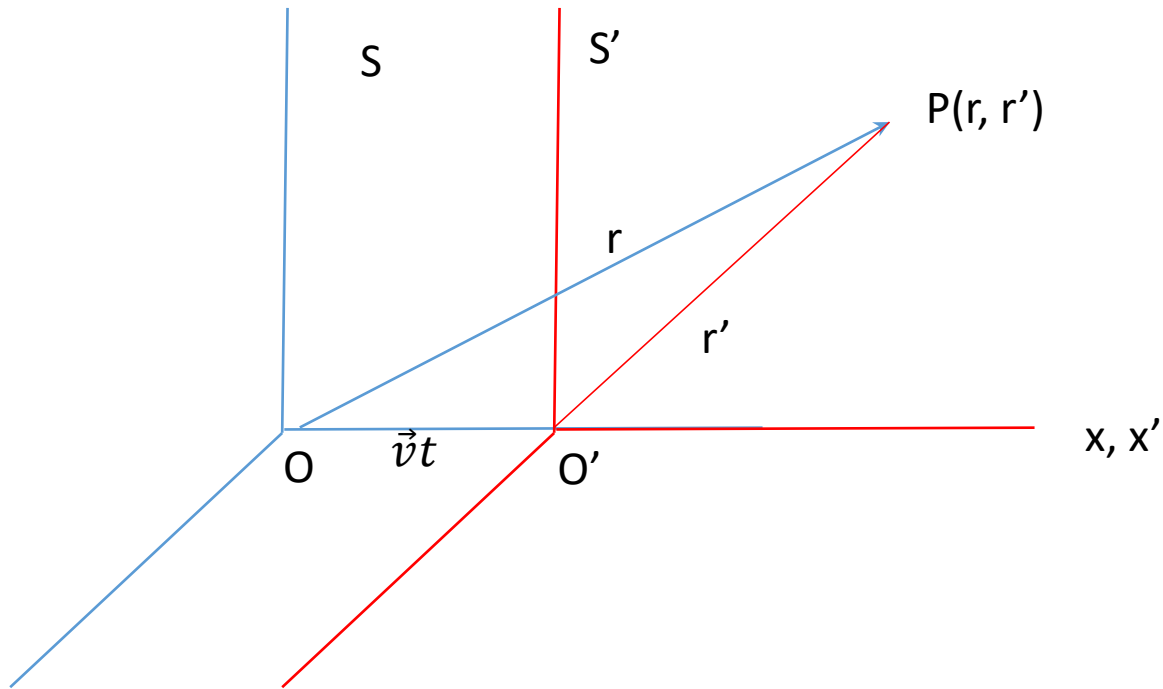
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Special Theory of Relativity

- **Reference Frame**
- **Newton's Law of Motion**
- **Natural state of motion**
- **Existence and Definition of real force**
- **Isolated Object and Inertial frame of reference**

Transformation between two inertial frame of references



$$\vec{r} = \vec{v}t + \vec{r}'$$

$$\dot{\vec{r}} = \vec{v} + \dot{\vec{r}'}$$

$$\vec{u} = \vec{v} + \vec{u}'$$

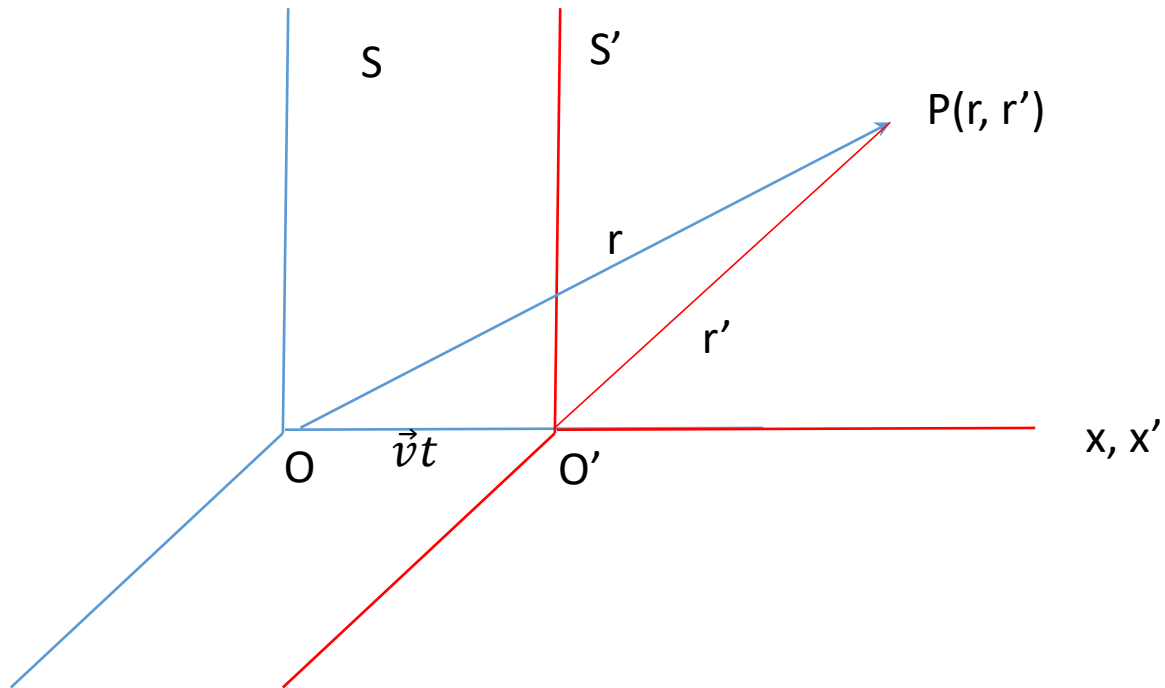
$$\dot{\vec{u}} = \dot{\vec{u}'}$$

$$\vec{a} = \vec{a}'$$

$$m\vec{a} = m\vec{a}'$$

$$\vec{F} = \vec{F}'$$

Galilean Transformation



$$\vec{r} = \vec{v}t + \vec{r}'$$
$$\vec{v} = (v_x, v_y, v_z)$$

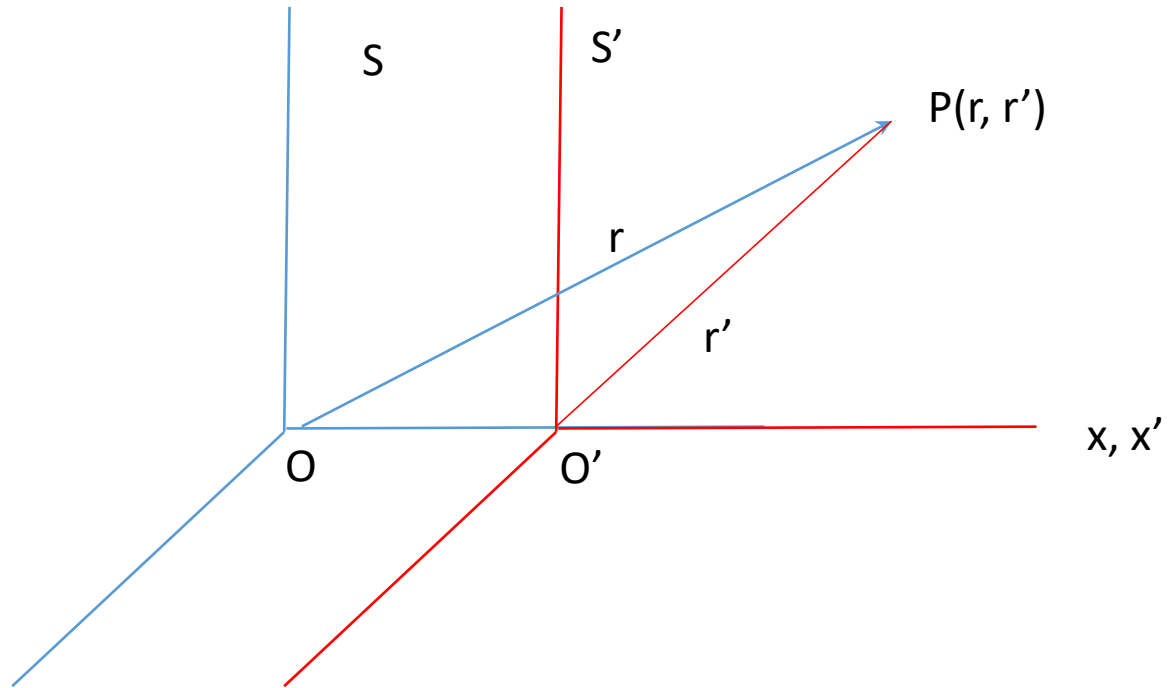
$$x = v_x t + x'$$

$$y = v_y t + y'$$

$$z = v_z t + z'$$

$$t = t'$$

Galilean Transformation

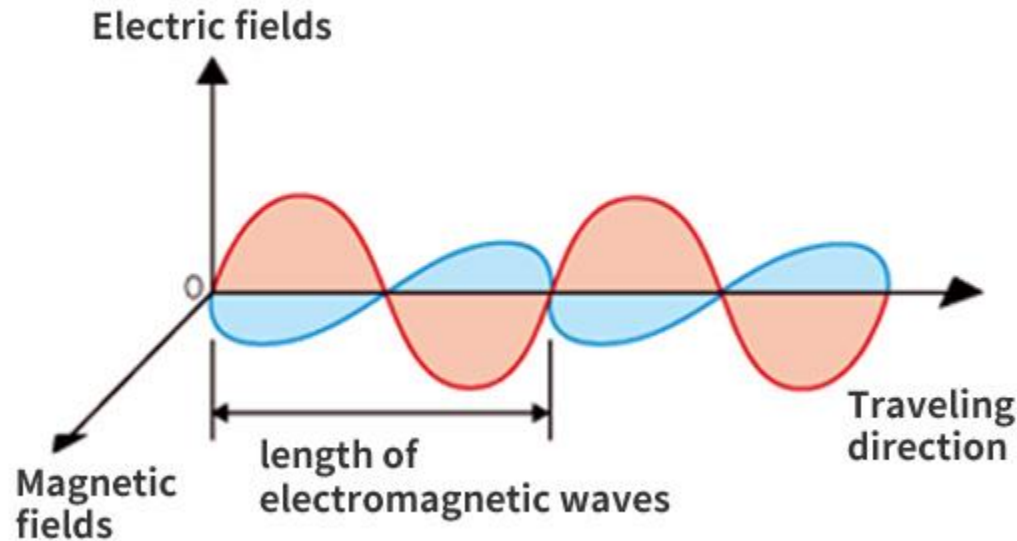


Lorentz Force:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$\vec{v}$ is frame dependent !!!

Electromagnetic radiation :



$$\frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} - \nabla^2 \vec{E} = 0$$

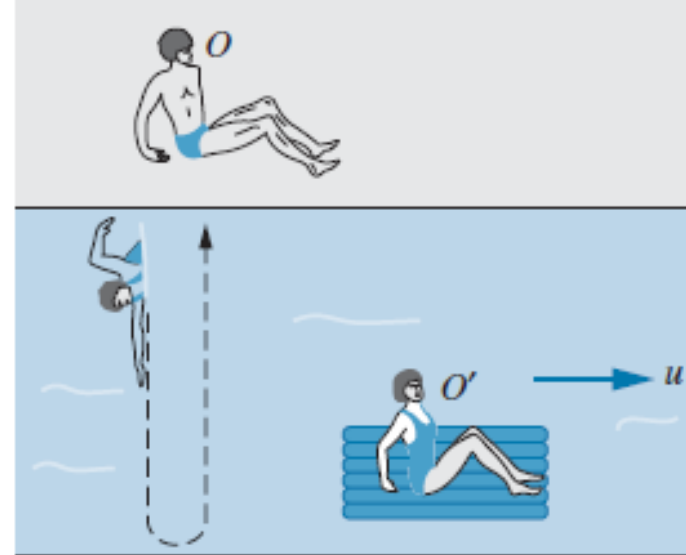
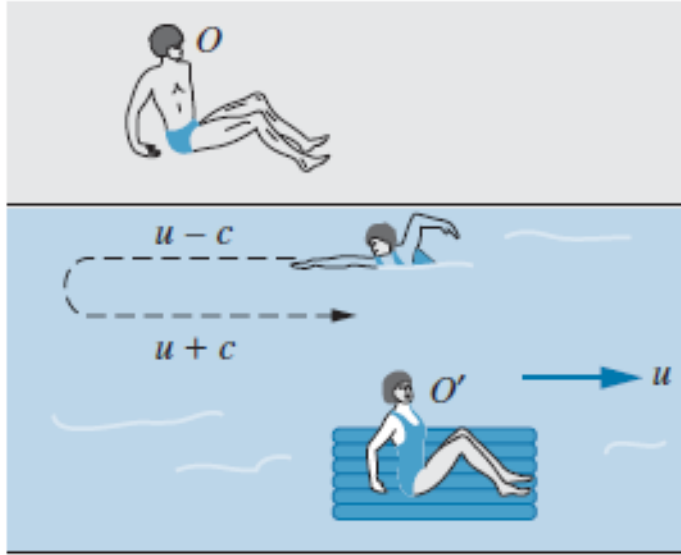
$$\frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} - \nabla^2 \vec{B} = 0$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 2.99792 \times 10^8 \text{ m/s}$$

Possible existence of Ether

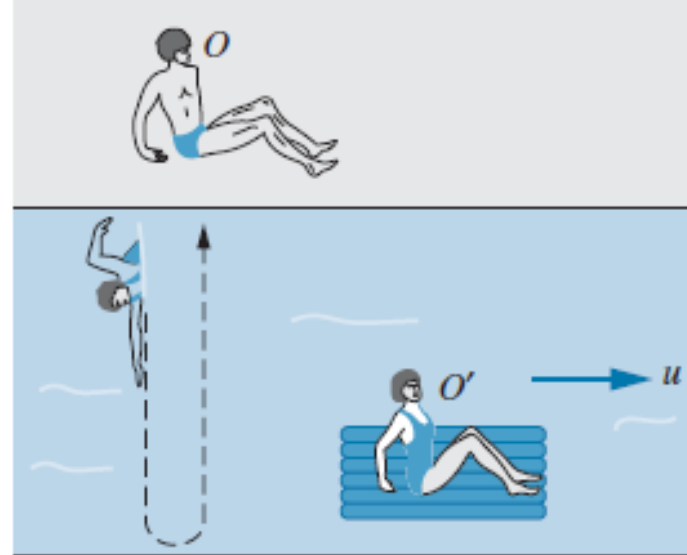
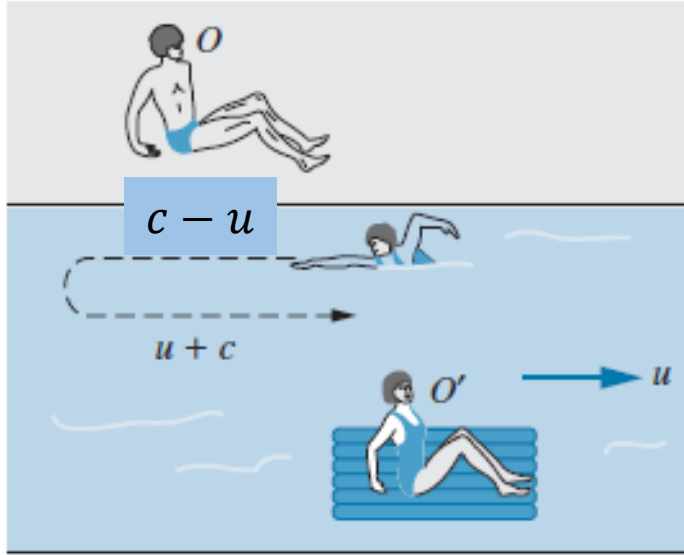
- **Velocity of light is only constant in some reference frame**
- **Comparing with the water in deep sea, space is filled with ether and the light travel with constant velocity c in this medium**
- **Without lose of generality we can take ether medium is the absolute rest frame**
- **We can then measure the absolute velocity (with respect to ether frame) for every inertial frame**
- **Earth will also have an absolute velocity and it can be measure by measuring velocity of light travelling in the different direction. !!**

Swimming in a river



A swimmer capable of swimming at a speed c in still water is swimming in a stream in which the current is u . Suppose the swimmer swims upstream a distance L and then returns downstream to the starting point. If the width of the river is also L and the swimmer make a round trip across the river. Find the time necessary to make the round trip, and compare it in two different situation.

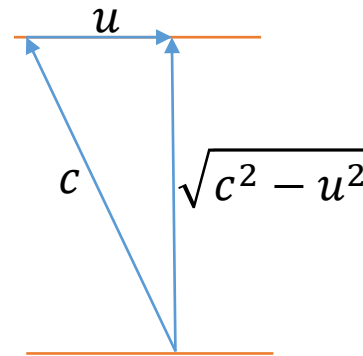
Swimming in a river



$$t_u = \frac{L}{c - u}$$

$$t_d = \frac{L}{c + u}$$

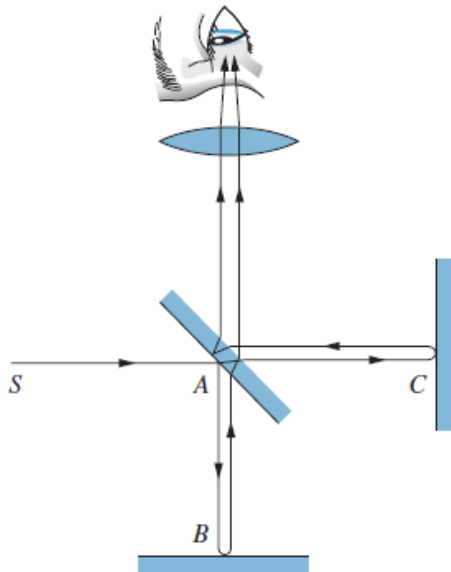
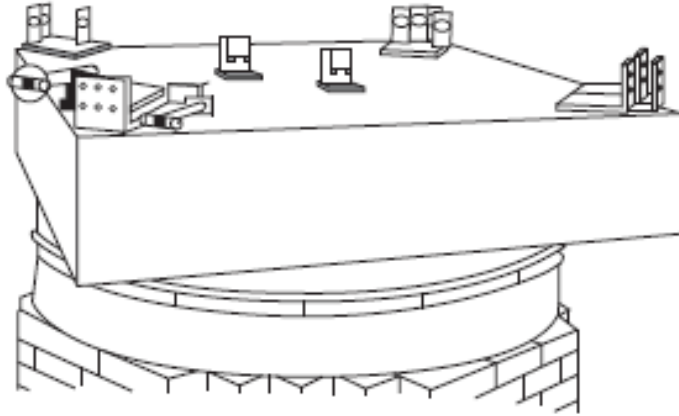
$$T_1 = t_u + t_d = \frac{L}{c - u} + \frac{L}{c + u} = \frac{2Lc}{c^2 - u^2}$$



$$T_2 = \frac{L}{\sqrt{c^2 - u^2}} \times 2$$

$$= \frac{2L}{\sqrt{c^2 - u^2}}$$

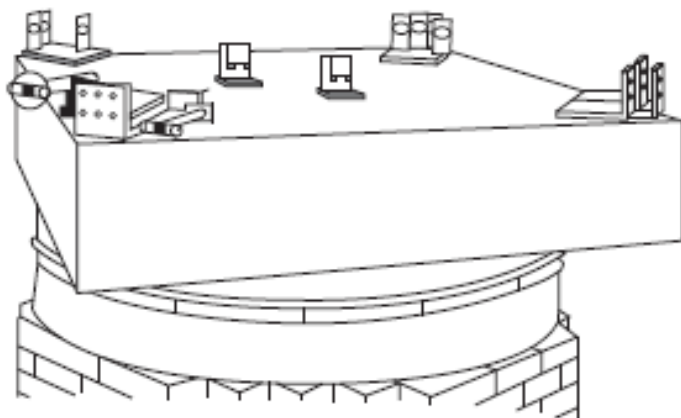
THE MICHELSON-MORLEY EXPERIMENT



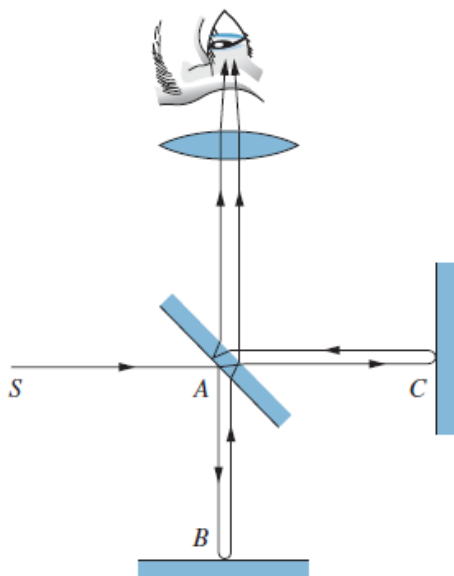
$$T_1 = \frac{2Lc}{c^2 - u^2} = \frac{2L}{c} \left(1 - \frac{u^2}{c^2} \right)^{-1} \\ = \frac{2L}{c} \left(1 + \frac{u^2}{c^2} \right)$$

$$T_2 = \frac{2L}{\sqrt{c^2 - u^2}} = \frac{2L}{c} \left(1 - \frac{u^2}{c^2} \right)^{-1/2} \\ = \frac{2L}{c} \left(1 + \frac{u^2}{2c^2} \right)$$

THE MICHELSON-MORLEY EXPERIMENT



$$\begin{aligned}\Delta T &= \frac{2L}{c} \left(1 + \frac{u^2}{c^2} \right) - \frac{2L}{c} \left(1 + \frac{u^2}{2c^2} \right) \\ &= \frac{Lu^2}{c^3}\end{aligned}$$



THE MICHELSON-MORLEY EXPERIMENT

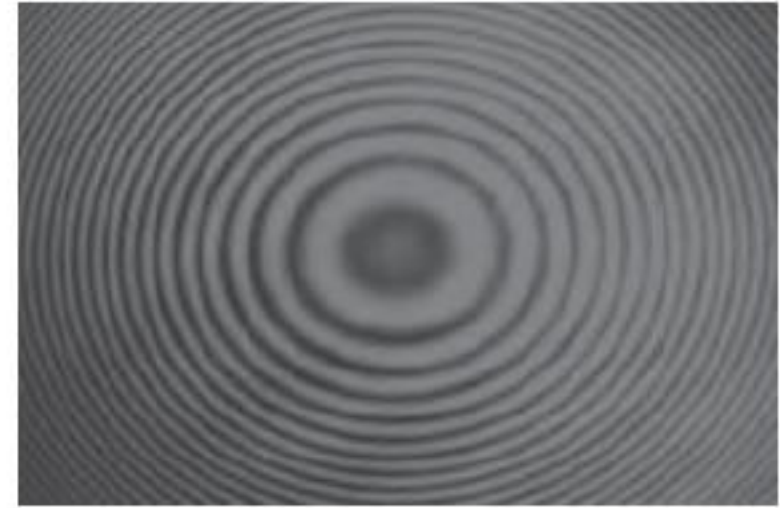
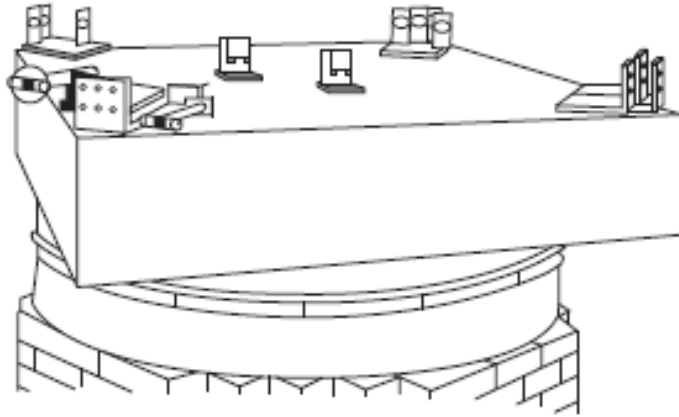
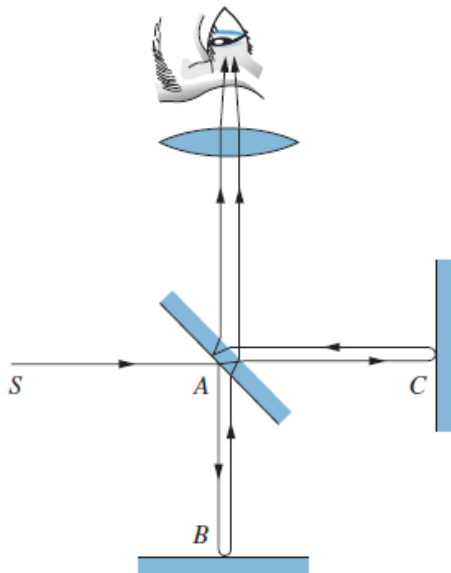


Fig. Interference fringes as observed with the Michelson interferometer



THE MICHELSON-MORLEY EXPERIMENT

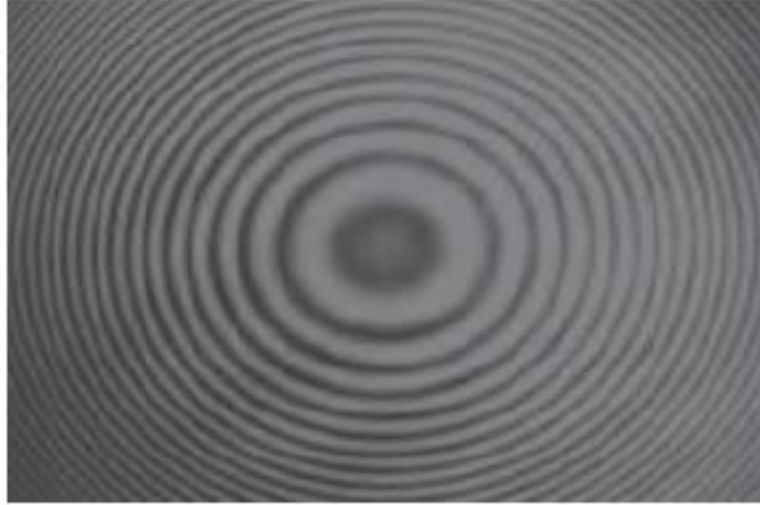


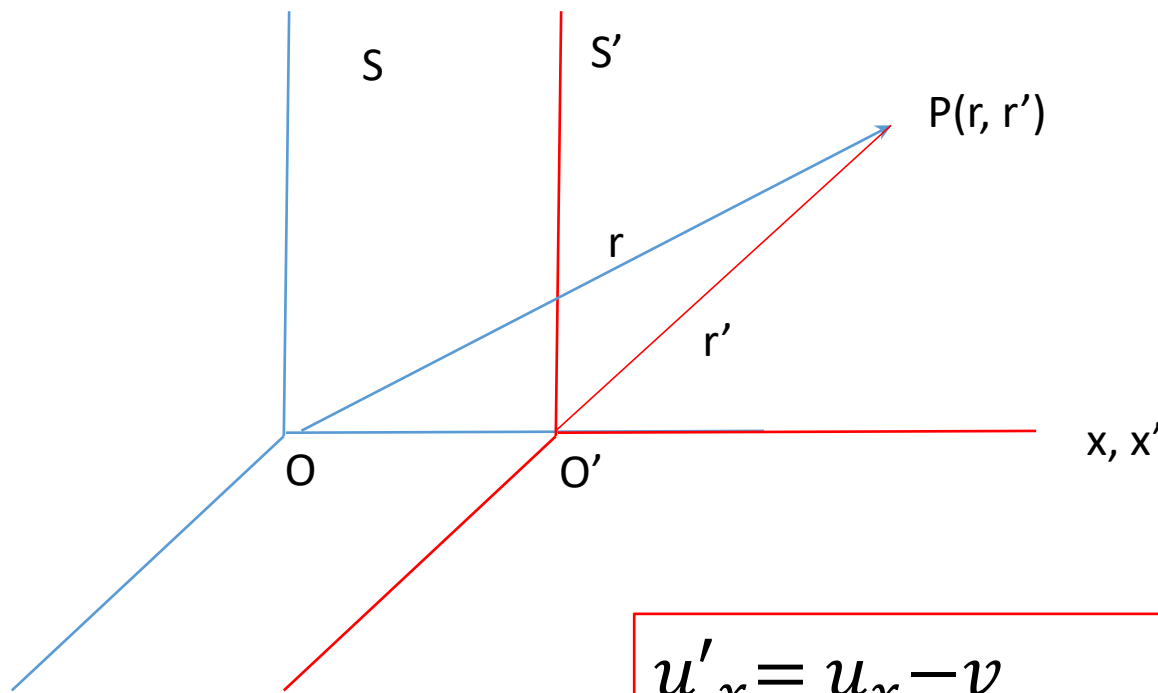
Fig. Interference fringes as observed with the Michelson interferometer

With a rotation of θ , Michelson interferometer setup

$$\Delta T = \frac{L\{u \cdot \cos(\theta)\}^2}{c^3}$$

- Michelson-Morley experiment confirms that there is no ether
- Light velocity does not change upon changing the reference frame

How to incorporate velocity of light in Galilean transformation rule??



$$\vec{r} = \vec{v}t + \vec{r}'$$

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

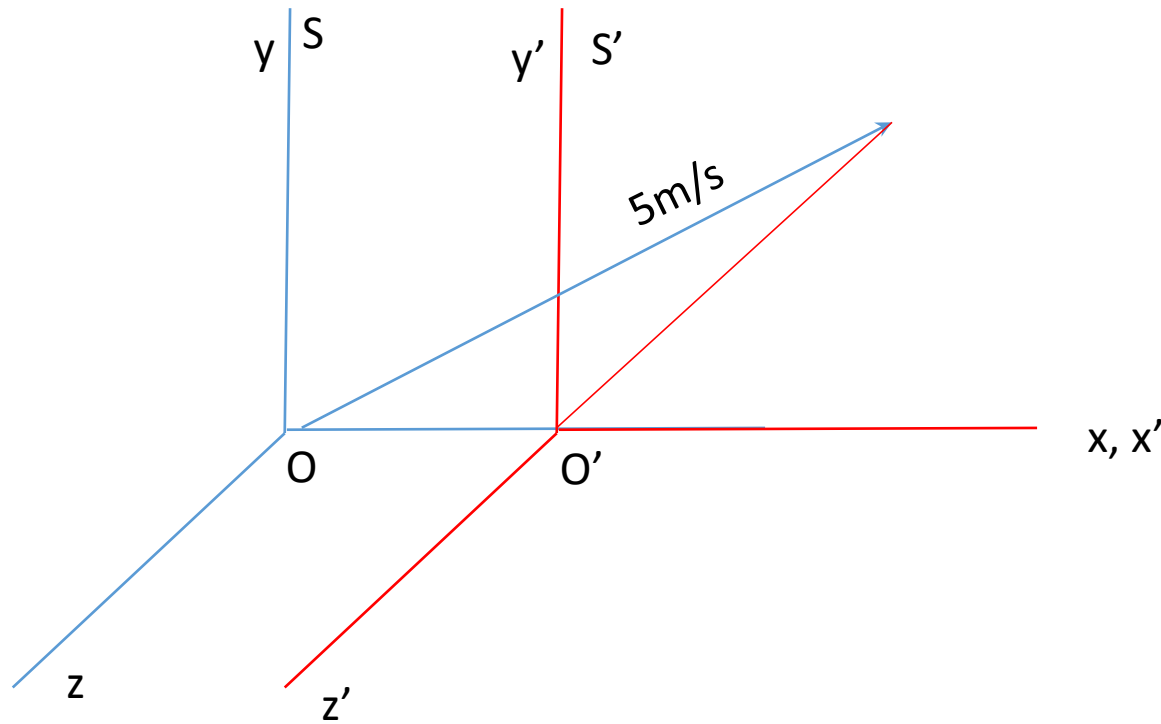
$$t' = t$$

$$u'_x = u_x - v$$

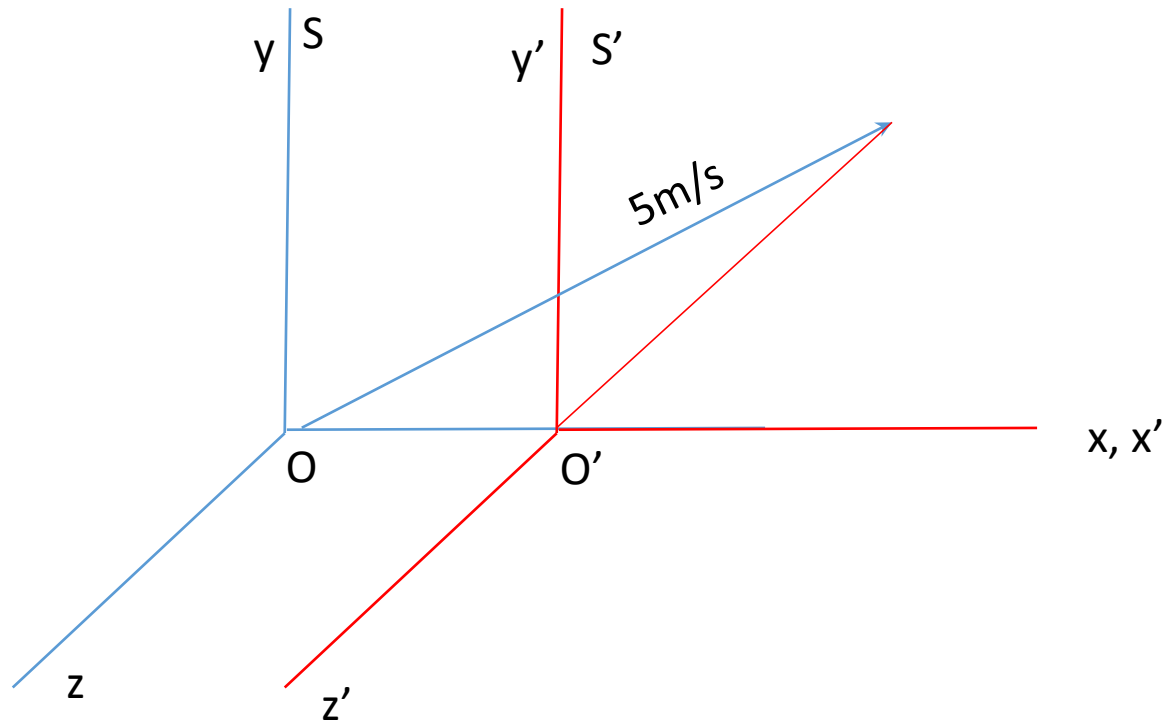
$$u'_y = u_y$$

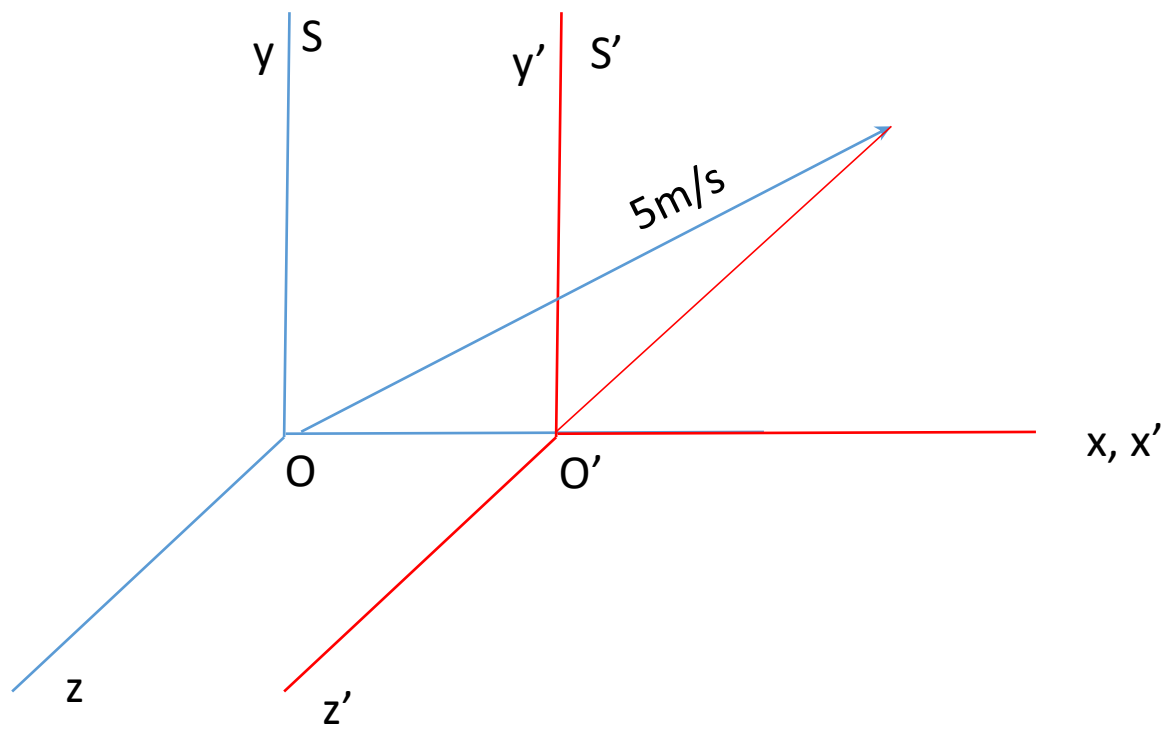
$$u'_z = u_z$$

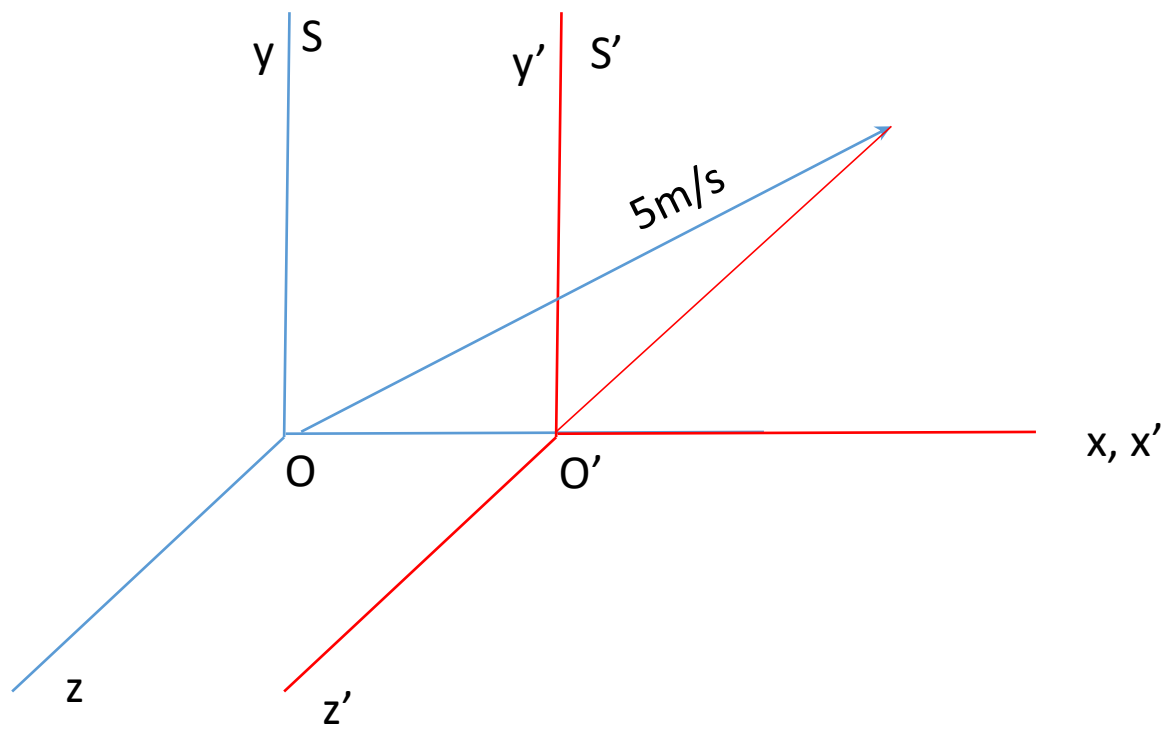
Example: An observer at S frame through a ball with a velocity 5 m/s in the xy-plane with an angle 30° . Another observer at frame S' moving with a velocity 2 m/s. with respect to S along x direction. What will be the position and velocity of the ball at time $t=2$ sec in different frames. Assume that both the origins of S and S' are coincide at time $t=0$



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Postulates of Special theory (of relativity)

1. Laws of physics are the same in all inertial reference frames.
2. The speed of light in a vacuum is the same (3.0×10^8 m/s) in all inertial reference frames regardless of the motion of the observer or source.

Conventional Galilean transformation rules need to be modified/change

One needs to look at time differently, which should transform in a very similar way as co-ordinate variables

Based on Einstein's postulate and physical observation, Lorentz proposed a set of transformation rules

Event :

An **event** is the instantaneous physical situation or occurrence associated with a point in spacetime, ie something happening in a point in space at an instance of time. For example: switching on lights. Exam starts. Ringing a bell, etc. these are denoted by $E(x,y,z,t)$

Simultaneous events:

More than one (two) events occurred at the same instance of time. ie. two event has same time co-ordinate.

Exmaple: exam starts at 10 am. , Train leave the station at 12 noon. etc.

Lorentz Transformation rules

Lorentz propose a set of transformation rule base on the observation, to accommodate Einstein's postulates. Note that there are no proof of this transformation, it is completely based on the physical observation (common sense !!!)

$$\begin{aligned}x' &= f_1(x, y, z, t), \\y' &= f_2(x, y, z, t) \\z' &= f_3(x, y, z, t) \\t' &= f_4(x, y, z, t)\end{aligned}$$

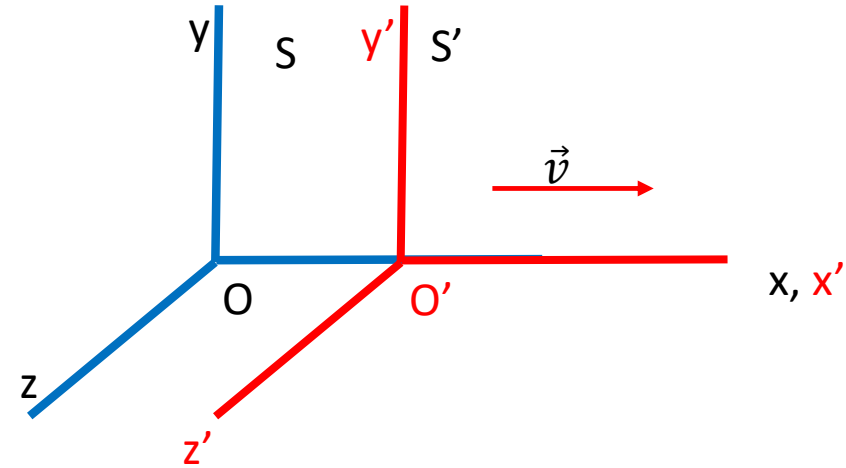
Since two inertial frame of reference is related with a constant velocity we may expect the transformation relation will be linear in nature

$$\begin{aligned}x' &= L_{xx}x + L_{xy}y + L_{xz}z + L_{xt}t + C1 \\y' &= L_{yx}x + L_{yy}y + L_{yz}z + L_{yt}t + C2 \\z' &= L_{zx}x + L_{zy}y + L_{zz}z + L_{zt}t + C3 \\t' &= L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t + C4\end{aligned}$$

All L_{ij} , C_i are constant we need to determine

Lorentz Transformation rules

$$\begin{aligned}x' &= L_{xx}x + L_{xy}y + L_{xz}z + L_{xt}t + C1 \\y' &= L_{yx}x + L_{yy}y + L_{yz}z + L_{yt}t + C2 \\z' &= L_{zx}x + L_{zy}y + L_{zz}z + L_{zt}t + C3 \\t' &= L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t + C4\end{aligned}$$



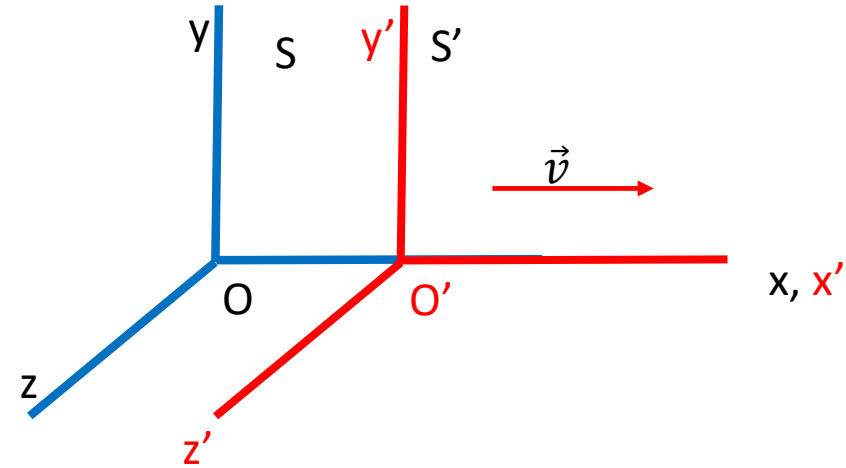
Let at $t = t' = 0$, both the origin coincide

i.e., when $x = y = z = t = 0$, then $x' = y' = z' = t' = 0$

➡ $C1 = C2 = C3 = C4 = 0$

Lorentz Transformation rules

$$\begin{aligned}x' &= L_{xx}x + L_{xy}y + L_{xz}z + L_{xt}t \\y' &= L_{yx}x + L_{yy}y + L_{yz}z + L_{yt}t \\z' &= L_{zx}x + L_{zy}y + L_{zz}z + L_{zt}t \\t' &= L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t\end{aligned}$$



Note that xy & $x'y'$ planes are coincides; For any event $z = 0, \rightarrow z' = 0$

➡ $L_{zx}x + L_{zy}y + L_{zt}t = 0$, for arbitrary values of x, y and z

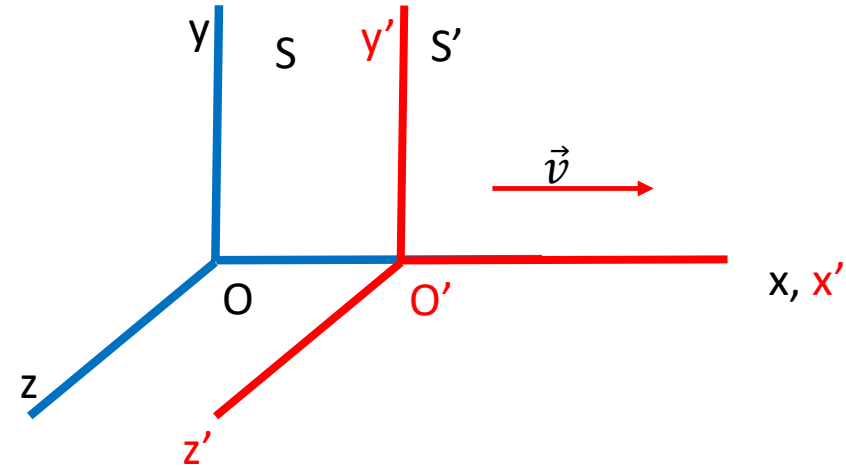
➡ $L_{zx} = L_{zy} = L_{zt} = 0$

Similarly xz & $x'z'$ planes are coincides; For any event $y = 0, \rightarrow y' = 0$

➡ $L_{yx} = L_{yz} = L_{yt} = 0$

Lorentz Transformation rules

$$\begin{aligned}x' &= L_{xx}x + L_{xy}y + L_{xz}z + L_{xt}t \\y' &= \cancel{L_{yx}}x + L_{yy}y + \cancel{L_{yz}}z + \cancel{L_{yt}}t \\z' &= \cancel{L_{zx}}x + \cancel{L_{zy}}y + L_{zz}z + \cancel{L_{zt}}t \\t' &= L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t\end{aligned}$$



Note that xy & $x'y'$ planes are coincides; For any event $z = 0, \rightarrow z' = 0$

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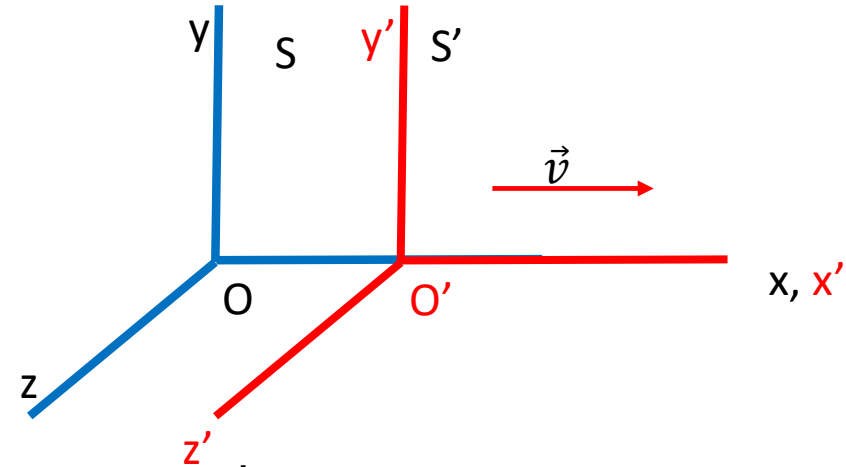
Lorentz Transformation rules

$$x' = L_{xx}x + L_{xy}y + L_{xz}z + L_{xt}t$$

$$y' = L_{yy}y$$

$$z' = L_{zz}z$$

$$t' = L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t$$



If an event occurs in the $y'z'$ plane ($x' = 0$) of S' frame at time t' , then x coordinate in S frame will be always vt

$$\longrightarrow 0 = L_{xx}vt + L_{xy}y + L_{xz}z + L_{xt}t = 0, \text{ for arbitrary values of } y, z \text{ and } t$$

$$\longrightarrow L_{xy} = L_{xz} = 0 \text{ \& } L_{xx}v + L_{xt} = 0 \rightarrow L_{xt} = -vL_{xx}$$

$$x' = L_{xx}(x - vt)$$

$$y' = L_{yy}y$$

$$z' = L_{zz}z$$

$$t' = L_{tx}x + L_{ty}y + L_{tz}z + L_{tt}t$$

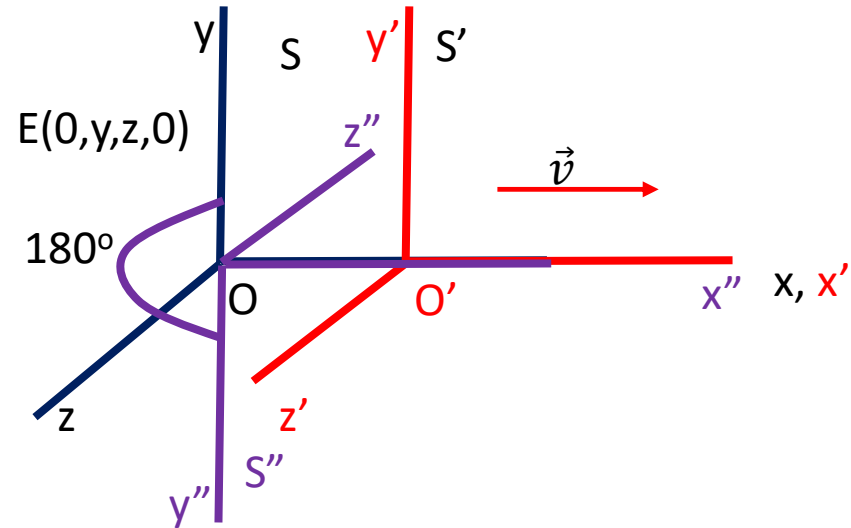
Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

$$y' = L_{yy}y$$

$$z' = L_{zz}z$$

$$t' = L_{tx}x + \cancel{L_{ty}y} + \cancel{L_{tz}z} + L_{tt}t$$



If an event occurs in the yz plane of S frame at time $t = 0$, i.e. $E(0, y, z, 0)$. The event point in the S'' frame will be $E''(0, -y, -z, t)$. Let us calculate the time of the same event in the S'

Transformation using S frame co-ordinate: $\longrightarrow t' = L_{ty}y + L_{tz}z$

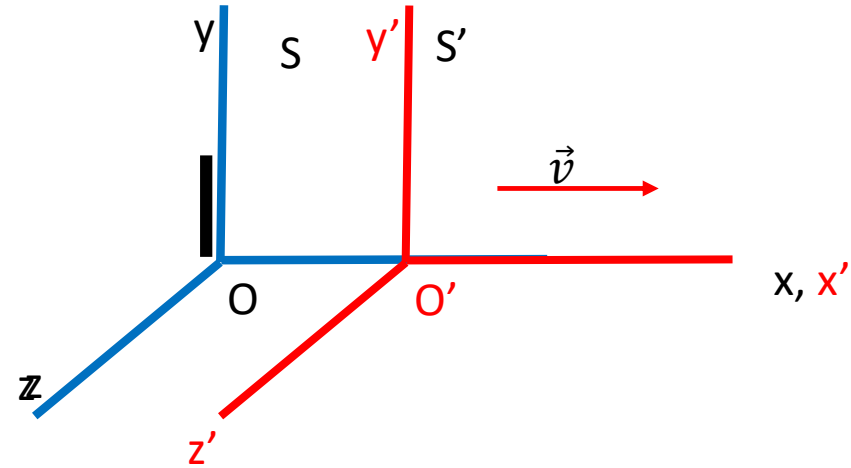
Transformation using S'' frame co-ordinate: $\longrightarrow t' = -L_{ty}y - L_{tz}z$

Measured time in S' should not depend on how an observer on other frame chooses his co-ordinate axis. $L_{tx} = L_{ty} = 0$

Lorentz Transformation rules

$$\begin{aligned}x' &= L_{xx}(x - vt) \\y' &= L_{yy}y \\z' &= L_{zz}z \\t' &= L_{tx}x + L_{tt}t\end{aligned}$$

Scale of length 1 placed at the origin along y-axis at $t=0$.
What is the length measure in S' ?

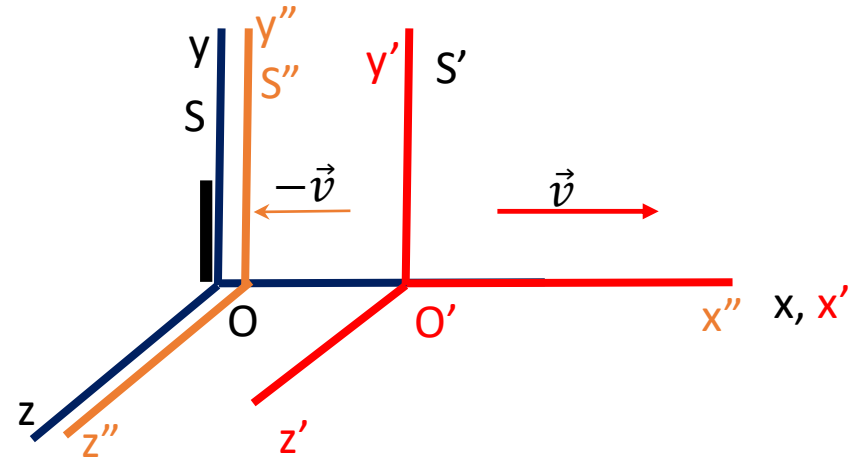


Distance between $E_1(0,0,0,0)$ & $E_2(0,1,0,0)$ is 1

Using transformation $E'_1(0,0,0,0)$ & $E'_2(0, L_{yy}, 0, 0)$, and the length is L_{yy}

Lorentz Transformation rules

$$\begin{aligned}x' &= L_{xx}(x - vt) \\y' &= L_{yy}y \\z' &= L_{zz}z \\t' &= L_{tx}x + L_{tt}t\end{aligned}$$



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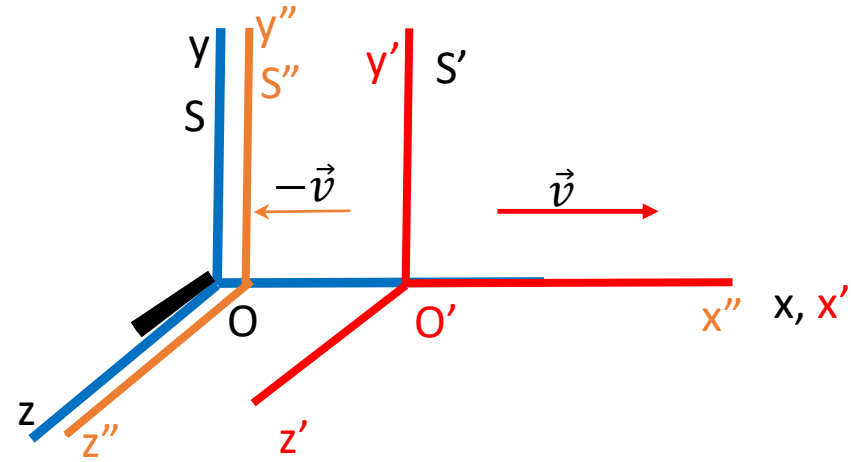
Let an observer at S'' moving with $\vec{v}$ along -ve x direction, with respect to S'

Using transformation from S' to S'' $E''_1(0,0,0,0)$ & $E''_2(0, L_{yy}^2, 0)$, and the length is L_{yy}^2

But S to S'' are the same frame $\rightarrow L_{yy}^2 = 1, \rightarrow L_{yy} = 1$

Lrentz Transformation rules

$$\begin{aligned}x' &= L_{xx}(x - vt) \\y' &= L_{yy}y \\z' &= L_{zz}z \\t' &= L_{tx}x + L_{tt}t\end{aligned}$$



Scale of length 1 placed at the origin along y-axis at $t=0$.
What is the length measure in S' ?

Distance between $E_1(0,0,0,0)$ & $E_2(0,1,0,0)$ is 1

Using transformation $E'_1(0,0,0,0)$ & $E'_2(0, L_{yy}, 0, 0)$, and the length is L_{yy}

Let an observer at S'' moving with $\vec{v}$ along -ve x axis with respect to S'

Using transformation from S' to S'' $E''_1(0,0,0,0)$ & $E''_2(0, L_{yy}^2, 0)$, and the length is L_{yy}^2

But S to S'' are the same frame $\rightarrow L_{yy}^2 = 1, \rightarrow L_{yy} = 1$

Placing the scale along the z axis and using similar argument we can prove that $L_{zz} = 1$

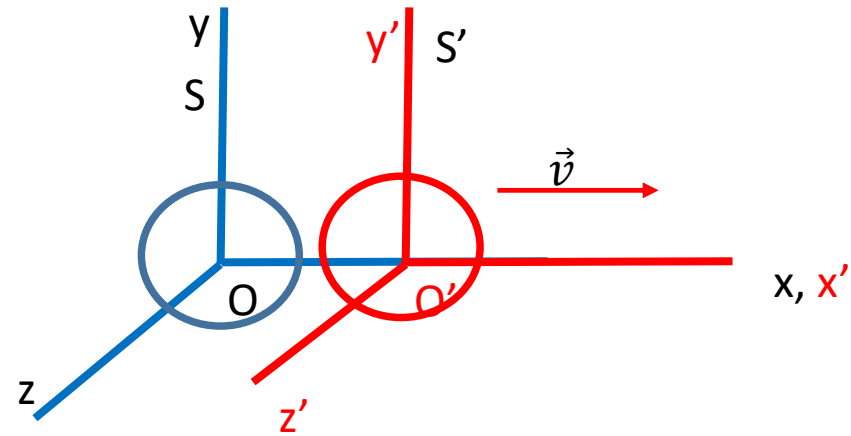
Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

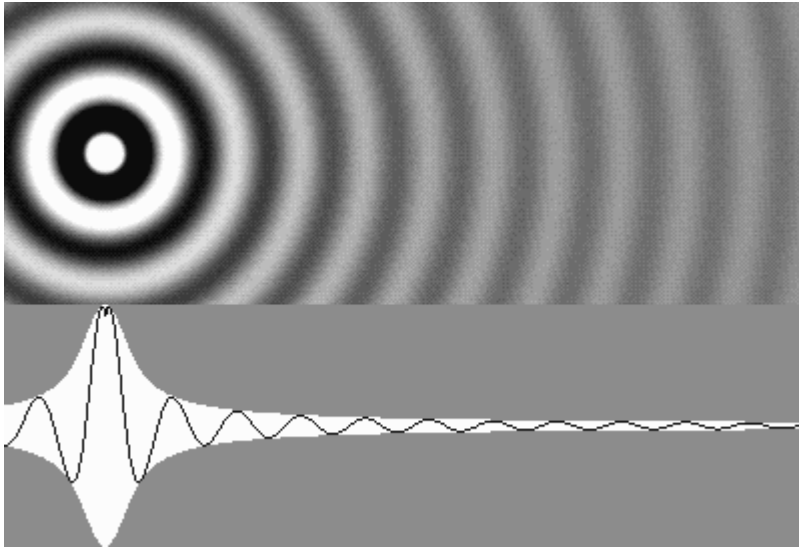
$$y' = y$$

$$z' = z$$

$$t' = L_{tx}x + L_{tt}t$$



We have not used the second postulate yet !!



Let at $t = t' = 0$, when origin of S and S' frame are coincide, a light flush

Wave fonts are moving with a velocity c , stays on the surface of the sphere of radius ct and ct' respectively to the observer in the S and S' frame.

The equation of the sphere in the frames are

$$x^2 + y^2 + z^2 = (ct)^2 \text{ and}$$

$$x'^2 + y'^2 + z'^2 = (ct')^2$$

Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = L_{tx}x + L_{tt}t$$

$$x^2 + y^2 + z^2 - (ct)^2 = 0$$

$$x'^2 + y'^2 + z'^2 - (ct')^2 = 0$$

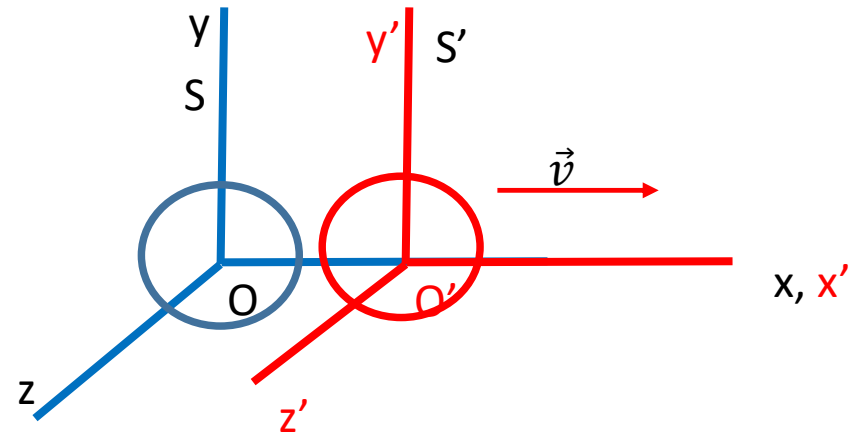
$$\{L_{xx}(x - vt)\}^2 + y^2 + z^2 - (c\{L_{tx}x + L_{tt}t\})^2 = 0$$

$$(L_{xx}^2 - c^2 L_{tx}^2) x^2 + y^2 + z^2 + (v^2 L_{xx}^2 - c^2 L_{tt}^2) t^2 - 2\{v L_{xx}^2 + c^2 L_{tx} L_{tt}\} xt = 0$$

$$L_{xx}^2 - c^2 L_{tx}^2 = 1$$

$$v^2 L_{xx}^2 - c^2 L_{tt}^2 = -c^2$$

$$2\{v L_{xx}^2 + c^2 L_{tx} L_{tt}\} = 0$$



Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = L_{tx}x + L_{tt}t$$

$$L_{xx}^2 - c^2 L_{tx}^2 = 1$$

$$v^2 L_{xx}^2 - c^2 L_{tt}^2 = -c^2$$

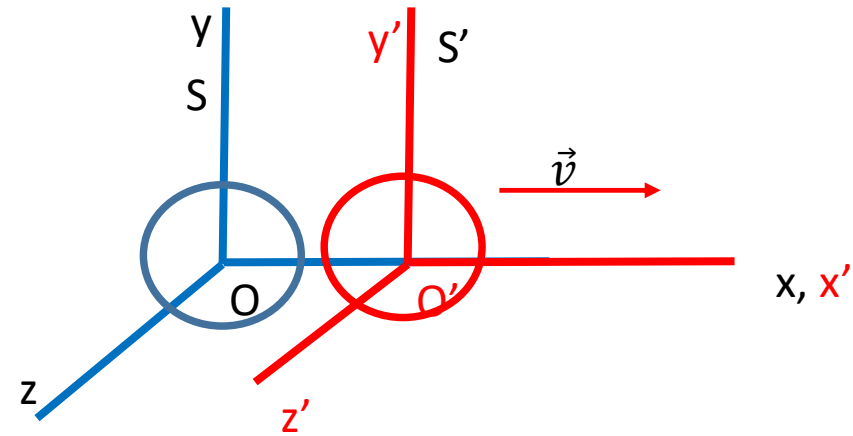
$$2\{v L_{xx}^2 + c^2 L_{tx} L_{tt}\} = 0$$

$$v L_{xx}^2 + c L_{tx} L_{tt} = 0$$

$$(v L_{xx}^2)^2 = (-c^2 L_{tx} L_{tt})^2 = (c^2 L_{tx}^2)(c^2 L_{tt}^2)$$

$$v^2 L_{xx}^4 = (L_{xx}^2 - 1)(c^2 + v^2 L_{xx}^2)$$

$$v^2 L_{xx}^4 = v^2 L_{xx}^4 + (c^2 - v^2) L_{xx}^2 - c^2$$



$$(c^2 - v^2) L_{xx}^2 = c^2$$

$$L_{xx}^2 = \frac{c^2}{(c^2 - v^2)}$$

$$L_{xx} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$$

Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = L_{tx}x + L_{tt}t$$

$$L_{xx}^2 - c^2 L_{tx}^2 = 1$$

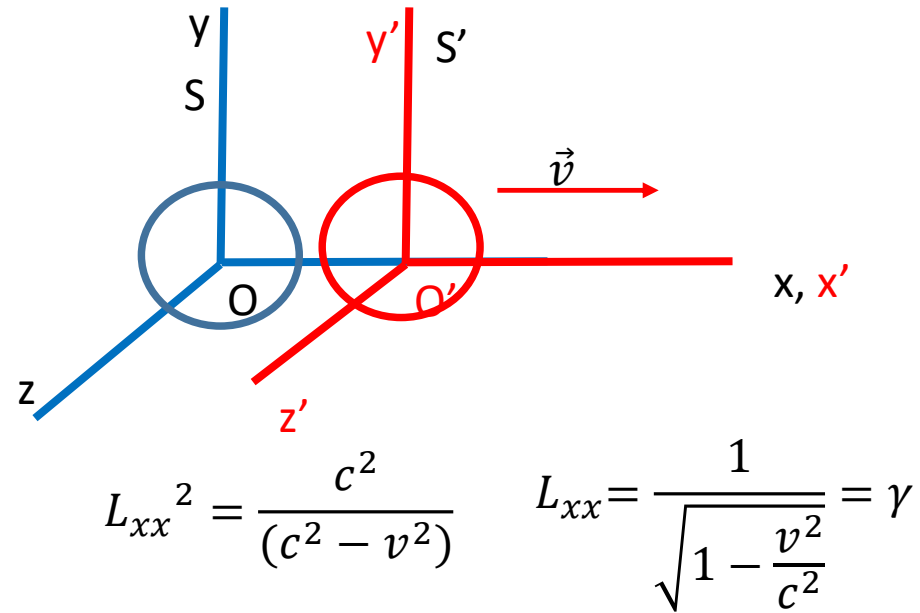
$$v^2 L_{xx}^2 - c^2 L_{tt}^2 = -c^2$$

$$2\{v L_{xx}^2 + c^2 L_{tx} L_{tt}\} = 0$$

$$\begin{aligned} c^2 L_{tt}^2 &= c^2 + v^2 L_{xx}^2 \\ &= c^2 + \frac{v^2 c^2}{(c^2 - v^2)} = \frac{c^4}{(c^2 - v^2)} \end{aligned}$$

$$L_{tt}^2 = \frac{c^2}{(c^2 - v^2)}$$

$$L_{tt} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$$



Lorentz Transformation rules

$$x' = L_{xx}(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = L_{tx}x + L_{tt}t$$

$$L_{xx}^2 - c^2 L_{tx}^2 = 1$$

$$v^2 L_{xx}^2 - c^2 L_{tt}^2 = -c^2$$

$$2\{v L_{xx}^2 + c^2 L_{tx} L_{tt}\} = 0$$

$$v\gamma^2 + c^2\gamma L_{tx} = 0 \rightarrow L_{tx} = -\frac{\gamma v}{c^2}$$

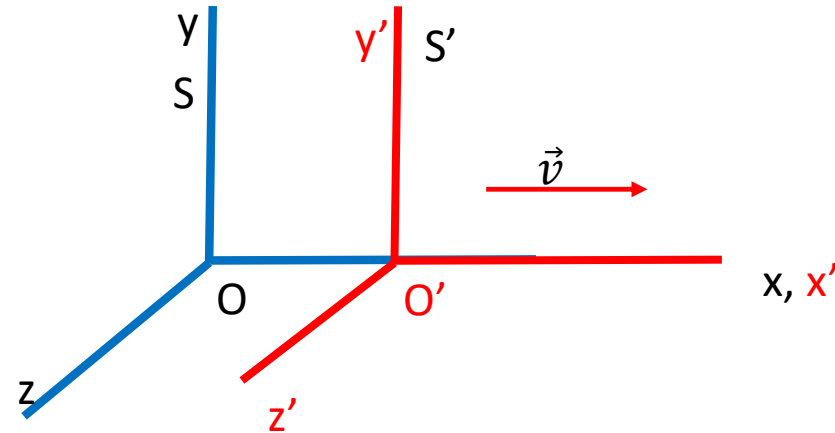
Lorentz Transformation

$$x' = \gamma(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$



$$L_{xx}^2 = \frac{c^2}{(c^2 - v^2)}$$

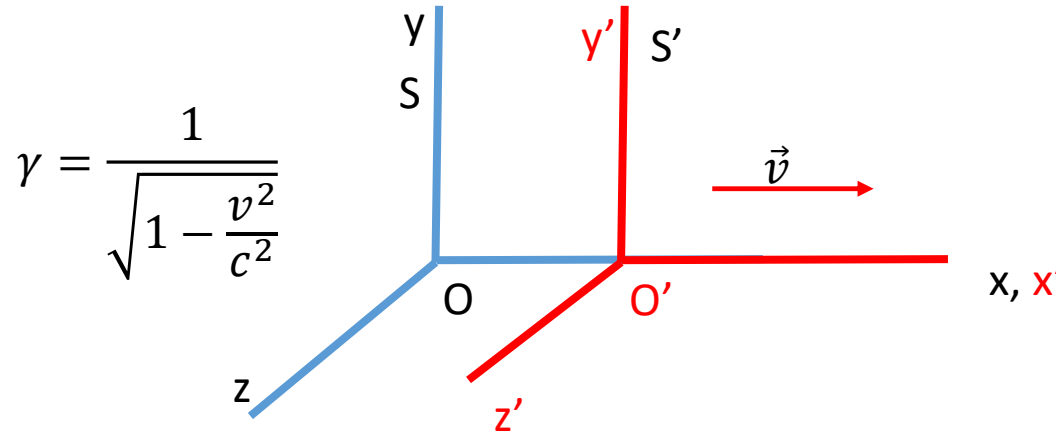
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Lorentz Transformation rules

$$\begin{aligned}x' &= \gamma(x - vt) \\y' &= y \\z' &= z \\t' &= \gamma\left(t - \frac{vx}{c^2}\right)\end{aligned}$$



Observation:

1. Co-ordinate parallel to the relative motion of the frame changes; which is expected
2. Co-ordinate perpendicular to the relative motion does not change; expected
3. The time of an event is frame dependent, is not same in different frame of reference !!!

What happened to the time interval between two events when we move from One reference frame to another ??

Proper time :

This is the time measured in a clock which is at rest with respect to the observer

Proper frame :

The frame of reference of the observer

Proper length :

Length of an object which is at rest with respect to the observer

Time dilation

Q. Suppose two events happened in the same place but in different time in S frame; Example a clock is at rest in the S frame and it stuck 1 pm and then it stuck 2 pm. What will be the time interval for an observer at S' frame which is moving at a constant velocity v along the +ve x direction?

Clearly the time interval measured in the S frame is proper time.

Let us define two event $E_1(x, t_1)$ and $E_2(x, t_2)$

The measure time interval in S is $(t_2 - t_1)$

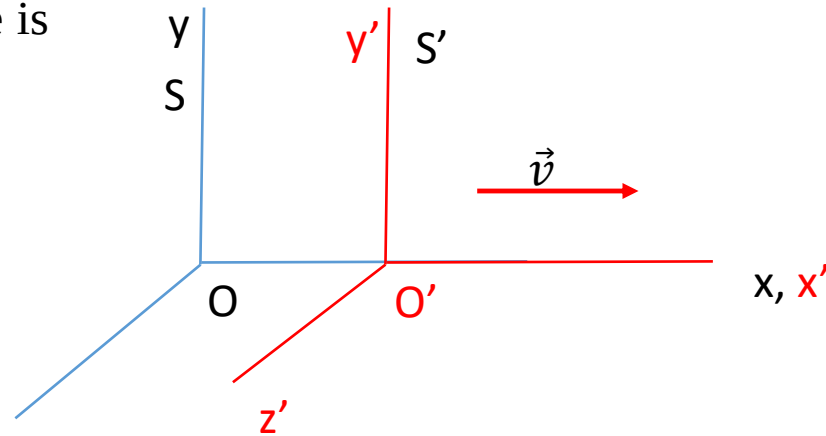
In the S' frame.

$$t'_1 = \gamma \left(t_1 - \frac{vx}{c^2} \right)$$

$$t'_2 = \gamma \left(t_2 - \frac{vx}{c^2} \right)$$

$$(t'_2 - t'_1) = \gamma(t_2 - t_1)$$

$$\text{Since } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} > 1 \quad \longrightarrow \quad (t'_2 - t'_1) > (t_2 - t_1)$$



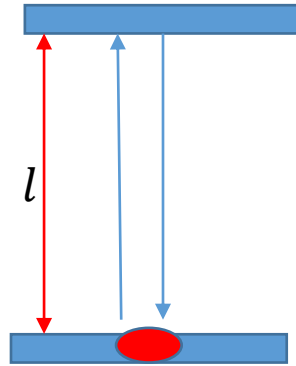
The one hour time interval in S is
Bigger time interval in S' frame.

So the clock is running slow for an observer at S' frame !!

Realization of Time dilation

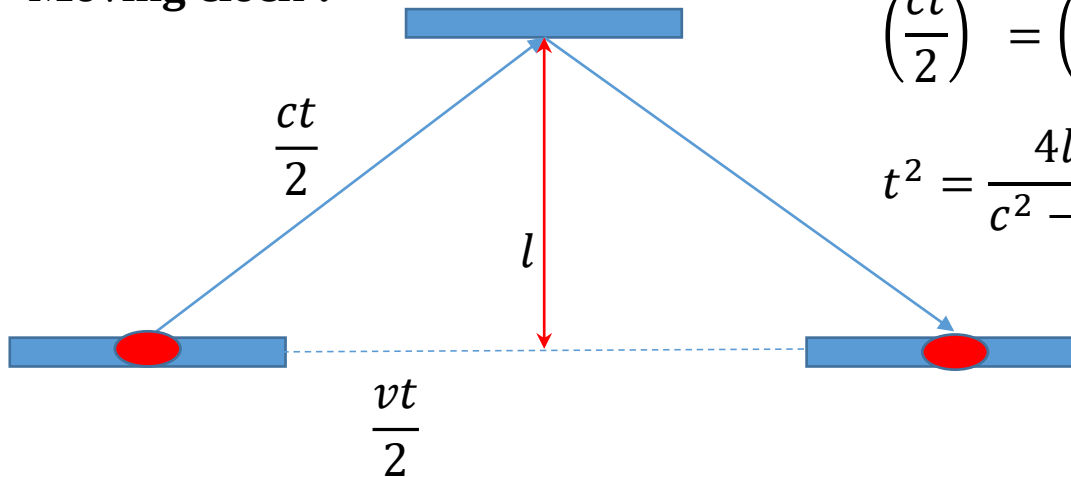
Construct the clock by using two mirror and a sensor.

Clock is at rest :



$$1 \text{ sec} = \frac{2l}{c}$$

Moving clock :



$$\left(\frac{ct}{2}\right)^2 = \left(\frac{vt}{2}\right)^2 + l^2$$

$$t^2 = \frac{4l^2}{c^2 - v^2} ;$$

$$t = \frac{\frac{2l}{c}}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} = \gamma \frac{2l}{c}$$

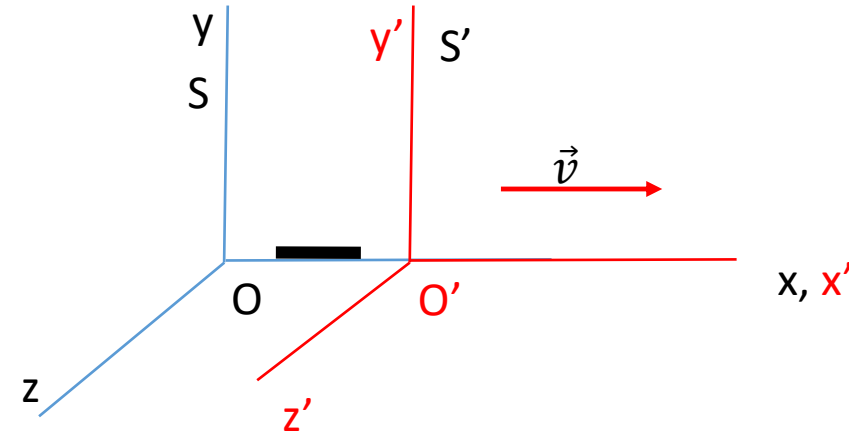
$$1 \text{ sec} = \gamma \frac{2l}{c} > \frac{2l}{c}$$

The clock running slow

Length contraction

Proper length : Length of an object which is at rest with respect to the observer

To measure the length of a moving object, observer has to measure the position of both the end at same time. (improper/measured length)



Let the length of an object l_0 rest at S frame and observer at S' frame trying to measure the length.

Let us define the event for the measurement $E'_1(x'_1, t')$ and $E'_2(x'_2, t')$

Note that S is moving with velocity $-v$ with respect to S'

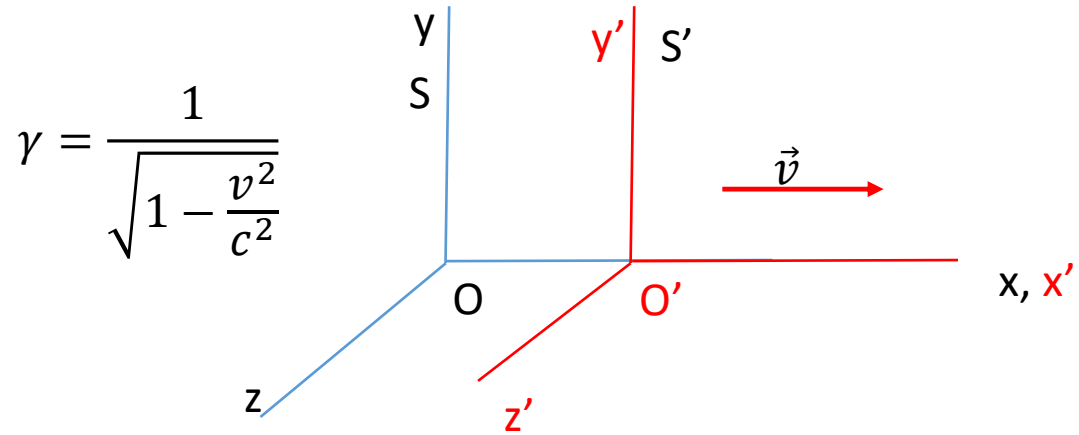
Using Lorentz transformation :

$$\begin{aligned} x_1 &= \gamma(x'_1 + vt') \\ x_2 &= \gamma(x'_2 + vt') \end{aligned} \quad \begin{aligned} (x_2 - x_1) &= \gamma(x'_2 - x'_1) \\ l_0 &= \gamma l \end{aligned} \quad \longrightarrow \quad l < l_0$$

Length of a moving object appears to be shorter than its original length

Lorentz Transformation rules

$$\begin{aligned}x' &= \gamma(x - vt) \\y' &= y \\z' &= z \\t' &= \gamma\left(t - \frac{vx}{c^2}\right)\end{aligned}$$



$$\begin{aligned}x &= \gamma(x' + vt') \\y &= y' \\z &= z' \\t &= \gamma\left(t' + \frac{vx'}{c^2}\right)\end{aligned}$$

Example 1: A rod of unit length in the xy-plane moving with a velocity $v=0.6c$ making an angle 30° with the x-axis. What will be the measure length and the inclined angle of the rod ?

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Example 2(B-5): Two observers, A on earth and B in a space-ship whose speed is 2×10^8 m/s. Both set their watches to the same time when the ship is abreast to the earth.

- (a) How much time must elapse by A's reckoning before the watches differ by 1 sec.
- (b) To A, B's watch seems to run slow. To B , how A's watch is running (faster, slower or same)?

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Example 3(B-7): How fast must a space-craft travel relative to the earth for each day on the spacecraft to correspond to two days on the earth?

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Example 4(B-17): An astronaut whose height on the earth is exactly 6ft, is lying parallel to the axis of a spacecraft moving at $0.9c$ relative to the earth. What is his height as measured by an observer in the same spacecraft ? By an observer on earth?

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Example 4(B-19): How much time does a meter stick moving at $0.1 c$ relative to an observer take to pass the observer? Consider the meter tick is parallel to its direction of motion.

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Example 4(B-19): Let S' is moving with a velocity $0.6c$ with respect to the frame S along $+x$ direction. One observer sitting at the origin of S observe an explosion at 20km away from him on the $+x$ axis and at the same time he also observe that the origin of S' is passing through him. What will the distance of explosion for an observer sitting at the origin of S' frame.

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Simultaneous events at two different place to an observer is no longer simultaneous to another observer in different reference frame